

Package ‘rbFoFR’

September 21, 2026

Type Package

Title Nonlinear Function-on-Function Regression from Fragments

Version 0.1.0

Description Estimation of the identifiable component of a nonlinear function-on-function regression operator when each predictor curve is observed only on a window of fixed length. The predictor is expanded in a known localized basis with independent symmetric scores, and the operator is a finite sum of products of scores (Walsh functionals) with function-valued coefficients. The package computes co-observation probabilities, the resolution order and the partition of interactions into estimated, trimmed and non-resolvable directions, and fits the resolvable coefficients by inverse-probability weighting or by a complete-case ratio estimator without reconstructing unobserved parts of the curves. Data generators for the seven simulation designs of the accompanying paper, Fisher information and risk benchmarks, and evaluation tools are included.

License GPL (≥ 2)

Encoding UTF-8

Depends R ($\geq 3.5.0$)

Imports stats, utils

NeedsCompilation no

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rbFoFR-package	<i>Nonlinear Function-on-Function Regression from Fragments</i>
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Description

Tools for nonlinear function-on-function regression when each predictor curve is observed only on a window $[A, A + \delta]$ of fixed length δ . The package separates the part of the regression operator that the observed data identify from the part that they cannot identify, and estimates the former without reconstructing unobserved parts of the curves.

Details

The model is $Y = G(X) + \varepsilon$, where the predictor is $X = \sum_{k=1}^K \xi_k \phi_k$. The ϕ_k are known smooth functions with pairwise disjoint supports I_k . The scores ξ_k are independent, have mean zero and variance one, and are symmetric about zero. The operator is a finite sum

$$G(X) = \sum_{J \subseteq \{1, \dots, K\}} g_J W_J(X), \quad W_J(X) = \prod_{k \in J} \xi_k,$$

where each coefficient g_J is a function in the response space. The set J is called an interaction of order $|J|$.

A window starting at a reveals the score ξ_k when it overlaps I_k on an interval of positive length. The co-observation probability of an interaction is $\rho_J = P\{J \subseteq K(A)\}$. The interaction J is resolvable when $\rho_J > 0$. The resolution order p is the largest number of scores that one window reveals.

The main results implemented are as follows.

- The conditional mean of Y given the fragment identifies exactly the coefficients g_J with $\rho_J > 0$, through $g_J = \rho_J^{-1} E[1\{J \subseteq K(A)\} Y W_J(X)]$.
- Pairwise disjoint interactions of order larger than p generate distinct operators with identical observed-data laws. This gives a minimax lower bound that does not decrease with the sample size.
- At the zero operator, the Fisher information is $\text{diag}(\rho_J) \otimes \Sigma^{-1}$. The inverse-probability weighted (IPW) estimator attains the local asymptotic minimax constant $\text{tr}(\Sigma) \sum_J \rho_J^{-1}$ on resolvable directions.

The main functions are grouped as follows.

Fitting: `rbfit` (Algorithm 2 of the paper) and `predict.rbfit`.

Data: `rbdata`, `gen_fofr`, `sim_design` and `design_setup` (the seven simulation designs).

Design geometry: `make_blocks`, `resolution_order`, `rho_set`, `classify_sets`, `band_measure` and `resolution_audit`.

Theory and evaluation: `fisher_info`, `ipw_risk`, `hypercube_check`, `parity_stat` and `evaluate_fit`.

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bases

*Predictor and Response Bases***Description**

phi_basis tabulates the localized orthonormal predictor functions $\phi_1, \dots, \phi_K$. response_basis tabulates an orthonormal response basis $e_1, \dots, e_d$.

Usage

```
phi_basis(blocks, sgrid)

response_basis(d, tgrid, tol = 1e-10)
```

Arguments

blocks	block system from make_blocks .
sgrid, tgrid	predictor and response grids.
d	number of response basis functions.
tol	tolerance for numerical degeneracy in the Gram–Schmidt step.

Details

Each ϕ_k is a normalized C^∞ bump, $\phi_k(s) \propto \exp\{-1/(1-z^2)\}$ for $|z| < 1$, where z maps I_k onto $(-1, 1)$. It is zero off I_k , positive inside, and has unit L^2 norm. Since the supports are disjoint, the system is orthonormal and $\langle X, \phi_k \rangle = \xi_k$. Positivity inside I_k is what allows a revealed score to be recovered from a partial view of its support.

The response basis starts with the constant function, followed by $\cos(2\pi t)$, $\sin(2\pi t)$, $\cos(4\pi t)$, and so on. The functions are orthonormalized with the trapezoidal inner product. For $d = 2$, this gives $e_1 = 1$ and $e_2 = \sqrt{2} \cos(2\pi t)$, as in the simulations.

Value

phi_basis: a length(sgrid) by K matrix. response_basis: a length(tgrid) by d matrix.

See Also

[make_blocks](#), [rbdata](#)

Examples

```
## Not run:
sgrid <- seq(0, 1, length.out = 301)
Phi <- phi_basis(make_blocks(), sgrid)
matplot(sgrid, Phi, type = "l", lty = 1)

E <- response_basis(2, seq(0, 1, length.out = 81))

## End(Not run)
```

Description

classify_sets computes ρ_J for every interaction up to a given order. It splits them into estimated, trimmed and non-resolvable directions. hypercube_check checks whether a family of interactions satisfies the conditions of the equal-law hypercube construction.

Usage

```
classify_sets(blocks, delta, qmax = 2, tau = 0.01)

hypercube_check(Jlist, blocks, delta)
```

Arguments

blocks	block system from make_blocks .
delta	window length.
qmax	largest interaction order.
tau	threshold on the co-observation probability.
Jlist	list of index sets.

Details

The conditional mean of the response given the fragment identifies exactly the coefficients g_J with $\rho_J > 0$. At the zero operator, the Fisher information over all 2^K directions has rank d times the number of resolvable sets. Its kernel consists exactly of the directions with $\rho_J = 0$. classify_sets returns this partition, with the resolvable directions further split at the threshold τ .

A direction with $\rho_J = 0$ is absent from the conditional mean. By itself, this does not make it invisible to the full observed law: overlapping hidden directions can leave parity information in the conditional covariance (see [parity_stat](#)).

A global impossibility result holds under the following conditions. The sets $J_1, \dots, J_m$ must be pairwise disjoint and share a common order $q > p$. Then, for any symmetric scores (Rademacher or Gaussian), the 2^m operators $G_\sigma = c \sum_i \sigma_i W_{J_i}$ have identical observed-data laws. Moreover, for every n ,

$$\max_{\sigma} E \|\hat{G} - G_\sigma\|^2 \geq \|c\|^2 m.$$

hypercube_check verifies these conditions.

Value

classify_sets: a list with the dictionary sets and its keys, the named vector rho, the keys of the estimated, trimmed and nonresolvable directions, tau, and the resolution order p.

hypercube_check: a list with valid (all conditions hold), m, q, p, disjoint and same_order.

See Also

[rho_set](#), [rbfit](#), [fisher_info](#)

Examples

```
## Not run:
blocks <- make_blocks()
cls <- classify_sets(blocks, delta = 0.22, qmax = 3, tau = 0.01)
cls$p
cls$estimated
length(cls$nonresolvable)

## design 2: the pair {1,2} is trimmed once tau exceeds 1/13
classify_sets(blocks, 0.22, qmax = 2, tau = 0.08)$trimmed

## design 7: the disjoint cubic family satisfies the hypercube conditions
hypercube_check(list(c(1, 2, 3), c(4, 5, 6)), blocks, delta = 0.22)$valid
## ... but not after the window grows to 0.38 (p becomes 3)
hypercube_check(list(c(1, 2, 3), c(4, 5, 6)), blocks, delta = 0.38)$valid

## End(Not run)
```

coobservation

Co-Observation Geometry

Description

Functions that describe what a window of length δ can observe jointly:

- the volume of the order- r co-observation band;
- the co-observation probability of a tuple of locations or of an index set;
- the resolution order p .

Usage

```
band_measure(r, delta)

rho_r(s, delta)

rho_set(J, blocks, delta)

resolution_order(blocks, delta)
```

Arguments

<code>r</code>	interaction order (may be a vector).
<code>delta</code>	window length in $[0, 1]$.
<code>s</code>	numeric vector of locations in $[0, 1]$.
<code>J</code>	index set, a vector of block indices.
<code>blocks</code>	block system from make_blocks .

Details

The band $B_r(\delta)$ contains the tuples of r locations whose range is at most δ . Its volume is

$$\lambda_r\{B_r(\delta)\} = r\delta^{r-1} - (r-1)\delta^r.$$

This decreases geometrically in r . For example, at $\delta = 0.2$, the values for orders 2 to 5 are 36%, 10.4%, 2.72% and 0.672%.

For window starts that are uniform on $[0, 1 - \delta]$, the probability that one window contains all of s is

$$\rho_r(s) = \frac{\{\min(s_{(1)}, 1 - \delta) - \max(s_{(r)} - \delta, 0)\}_+}{1 - \delta}.$$

For $r \geq 2$, $1/\rho_r$ is not integrable over the band, which is why the estimators use a positive threshold.

A window starting at a reveals block k when $a \in (l_k - \delta, u_k)$. So

$$\rho_J = \frac{|(\max_{k \in J}(l_k - \delta), \min_{k \in J} u_k) \cap [0, 1 - \delta]|}{1 - \delta},$$

computed exactly, and $\rho_\emptyset = 1$.

The resolution order $p = \sup_a |K(a)|$ is the largest number of blocks revealed by one window. It is computed exactly by sweeping the interval endpoints. Every interaction of order larger than p is non-resolvable.

Value

band_measure: the band volume. rho_r and rho_set: a probability. resolution_order: an integer.

See Also

[classify_sets](#), [resolution_audit](#)

Examples

```
## Not run:
## band volumes at delta = 0.2
round(band_measure(2:5, 0.2), 4)

rho_r(c(0.30, 0.40), delta = 0.22)

blocks <- make_blocks()
rho_set(c(1, 2), blocks, 0.22) # 1/13, the rare pair of design 2
rho_set(c(1, 2, 3), blocks, 0.22) # 0: non-resolvable
resolution_order(blocks, 0.22) # p = 2
resolution_order(blocks, 0.38) # p = 3

## End(Not run)
```

Description

Returns the block system, grids, error covariance, score distribution, true operator and tuning values of each simulation design in the paper.

Usage

```
design_setup(design, delta = NULL)
```

Arguments

design integer from 1 to 7.
delta window length; used only in design 3, where it defaults to 0.22.

Details

Designs 1–4, 6 and 7 share the following settings:

- $K = 6$ bumps of width 0.03 centred at 0.045, 0.105, 0.415, 0.475, 0.815 and 0.875;
- 301- and 81-point grids;
- $d = 2$, with error covariance Σ having variances 0.60 and 0.40 and covariance 0.15;
- window length $\delta = 0.22$.

Coefficient rows are listed in the order of the index sets.

Design 1 (resolvable nonlinear effects). Rademacher scores. The active sets are $\{3\}$, $\{4\}$, $\{3,4\}$, $\{5,6\}$ and $\{1,2\}$, with rows (1,-0.4), (-0.7,0.5), (0.9,0.3), (-0.6,-0.8) and (0.5,0.2). Every active direction has $\rho_J > 0$. Uses $q_{\max} = 2$ and $n \in \{200, 500, 1000\}$.

Design 2 (thresholding a rare interaction). The active sets are $\{3\}$, $\{3,4\}$, $\{5,6\}$ and $\{1,2\}$, with rows (0.70,-0.25), (0.75,0.25), (-0.55,-0.65) and (0.95,0.35). Here $\rho_{\{1,2\}} = 1/13$, so thresholds 0.08 and 0.10 discard $\{1,2\}$ and add squared bias 1.025. Uses $\tau \in \{0.005, 0.025, 0.05, 0.08, 0.10\}$ (in `extra$tau_grid`) and $n \in \{100, 250, 500, 1000\}$.

Design 3 (changing resolution). Uses the design 1 operator plus $\gamma_{\{1,2,3\}} = c$ and $\gamma_{\{4,5,6\}} = -c$, with $c = (1, -0.5)$. Uses $q_{\max} = 3$, $\delta \in \{0.12, 0.22, 0.30, 0.38, 0.46, 1\}$ (in `extra$delta_grid`) and $n \in \{300, 1000\}$. For $\delta \leq 0.30$, the two cubic terms are non-resolvable and contribute omitted energy 2.5.

Design 4 (Gaussian scores). Uses the design 3 operator at $\delta = 0.22$ with standard normal scores, and $n \in \{250, 500, 1000, 2000\}$.

Design 5 (full observation). $\delta = 1$, $K = 4$ bumps of width 0.18 centred at 0.14, 0.38, 0.62 and 0.86, Gaussian scores, 101- and 61-point grids, and $\Sigma = \text{diag}(0.20, 0.16)$. Singleton effects with rows (0.65,0.15), (-0.55,0.20), (0.45,-0.15) and (-0.35,-0.10). Uses $q_{\max} = 1$ and $n \in \{100, 250, 500\}$.

Design 6 (local information). Uses the five design 1 directions. The operator is zero or $H/\sqrt{n}$, where H (component H) has rows (0.70,-0.30), (-0.55,0.35), (0.65,0.20), (-0.45,-0.55) and (0.55,0.15). Uses $n \in \{200, 400, 800, 1600\}$. The information constant is $\text{tr}(V) = 28.917$.

Design 7 (equal laws and visible parity). The hypercube family $c(\sigma_1 W_{\{1,2,3\}} + \sigma_2 W_{\{4,5,6\}})$ and the overlapping family $c(\sigma_1 W_{\{1,3\}} + \sigma_2 W_{\{1,2,3\}})$, with $c = (1, -0.5)$. Both families are listed in `extra`.

Value

A list with components design, blocks, delta, sgrid, tgrid, Sigma, scores, J (true index sets), Gamma (true coefficients), H (design 6 only), qmax, tau and extra (design-specific grids and families).

See Also

[sim_design](#)

Examples

```
## Not run:
st <- design_setup(3)
st$J
st$extra$delta_grid

## geometry of design 3 across window lengths
for (dl in st$extra$delta_grid) {
  rho <- sapply(st$J, rho_set, blocks = st$blocks, delta = dl)
  cat("delta =", dl, " p =", resolution_order(st$blocks, dl),
      " omitted energy =", sum(rowSums(st$Gamma^2)[rho == 0]), "\n")
}

## End(Not run)
```

evaluation

Operator and Prediction Errors

Description

Error measures that separate operator recovery from prediction from a fragment.

Usage

```
evaluate_fit(fit, test, Jtrue, Gtrue)

operator_error(fit, Jtrue, Gtrue)

truth_matrix(sets, Jtrue, Gtrue)

prediction_error(Yhat, Y)

curve_mise(pred, truth, grid)
```

Arguments

fit	an object of class "rbfit".
test	an "rbdata" object with independent test subjects.
Jtrue, Gtrue	true index sets and coefficient matrix, for example setup\$J and setup\$Gamma of sim_design .
sets	list of index sets of a dictionary.

<code>Yhat, Y</code>	matrices of predicted and observed response coordinates.
<code>pred, truth</code>	matrices of predicted and true curves, one row per subject.
<code>grid</code>	grid on which the curves are tabulated.

Details

Because the Walsh functionals and the response basis are orthonormal, the operator loss is $L_{op} = \|\hat{G} - G\|_{L^2(P_X)}^2 = \sum_J \|\hat{\gamma}_J - \gamma_J\|^2$. Coefficients are extended by zero outside the dictionary. `operator_error` splits this loss into three parts:

- the estimated directions ($\rho_J \geq \tau$);
- the trimmed directions ($0 < \rho_J < \tau$);
- the non-resolvable directions ($\rho_J = 0$).

For the truncated estimators, the last two parts are exactly the threshold bias and the energy absent from the conditional mean.

`evaluate_fit` also computes the fragment-signal loss $L_{sig} = N^{-1} \sum_i \|\hat{m}_i - m_i\|^2$. Here $m_i = \sum_J \gamma_J Z_{iJ}$ is the true signal that is visible in test subject i 's fragment. It also computes the response loss and the response loss relative to the mean predictor. A low fragment-signal loss can coexist with a large operator loss.

`curve_mise` is the mean integrated squared error between curves, computed by the trapezoidal rule. It is used in design 5.

Value

`evaluate_fit`: a named vector with `operator_total`, `operator_estimated`, `operator_threshold`, `operator_nonresolvable`, `fragment_signal_mse`, `response_mse` and `response_relative`.

`operator_error`: a list with `total`, `estimated`, `threshold`, `nonresolvable`, `nonresolvable_truth` (the true energy of the non-resolvable directions) and `per_set`.

`truth_matrix`: the true coefficients aligned with sets.

`prediction_error`: a list with `mse`, `baseline` and `relative`.

`curve_mise`: a number.

See Also

[rbfit](#), [predict.rbfit](#)

Examples

```
## Not run:
## Design 3 at delta = 0.22: large operator loss, small fragment loss
R <- 20
res <- t(replicate(R, {
  tr <- sim_design(3, n = 1000, delta = 0.22)
  te <- sim_design(3, n = 1000, delta = 0.22)
  f <- rbfit(tr, qmax = 3, tau = 0.01)
  evaluate_fit(f, te, tr$setup$J, tr$setup$Gamma)
}))
colMeans(res)[c("operator_total", "operator_nonresolvable",
               "fragment_signal_mse")]

## Design 1: error of each direction
```

```

dat <- sim_design(1, n = 1000, seed = 1)
fit <- rbfit(dat, qmax = 2)
round(operator_error(fit, dat$setup$J, dat$setup$Gamma)$per_set, 4)

## End(Not run)

```

gen_fofr

*Generate Data from the Localized-Score Model***Description**

Simulates fragmented predictors and response curves from the finite multilinear model with localized predictor basis and Gaussian errors.

Usage

```

gen_fofr(n, blocks, delta, sgrid, tgrid, Jlist, Gamma, Sigma,
        scores = c("rademacher", "gaussian"),
        Phi = NULL, E = NULL, seed = NULL)

```

Arguments

n	sample size.
blocks	block system from make_blocks .
delta	window length in $(0, 1]$.
sgrid, tgrid	predictor and response grids.
Jlist	list of index sets of the active interactions; use <code>integer(0)</code> for the intercept.
Gamma	coefficient matrix with one row per element of Jlist and d columns.
Sigma	$d \times d$ covariance matrix of the error coordinates.
scores	distribution of the scores: "rademacher" (± 1 with probability $1/2$) or "gaussian" (standard normal).
Phi, E	optional predictor and response bases.
seed	optional random seed.

Details

The data are generated from

$$X_i = \sum_{k=1}^K \xi_{ik} \phi_k, \quad Y_i^* = \sum_j \gamma_{J_j} W_{J_j}(X_i) + \varepsilon_i, \quad Y_i(t) = e(t)^\top Y_i^*,$$

where $\varepsilon_i \sim N_d(0, \Sigma)$. Windows start at $A_i \sim U[0, 1 - \delta]$, independently of the scores, and $A_i = 0$ when $\delta = 1$.

With Gaussian scores, X is a Gaussian process with C^∞ paths. The resolution barrier holds for any symmetric scores, so it is not an artifact of the Rademacher case.

The latent quantities are returned in truth for evaluation only. The estimators never use them.

Value

An object of class "rbdata" with an extra component `truth`, a list containing the scores `xi`, the full curves `Xfull`, the Walsh features `W`, the conditional mean coordinates `mu`, the response coordinates `Ystar`, and `Jlist`, `Gamma`, `Sigma` and scores.

See Also

[sim_design](#), [rbdata](#)

Examples

```
## Not run:
## a design of your own: one resolvable pair and one hidden triple
blocks <- make_blocks()
sgrid <- seq(0, 1, length.out = 301)
tgrid <- seq(0, 1, length.out = 81)
J <- list(c(3, 4), c(1, 2, 3))
G <- rbind(c(1, 0.5), c(0.8, -0.4))
dat <- gen_fofr(500, blocks, delta = 0.22, sgrid, tgrid, J, G,
               Sigma = diag(c(0.6, 0.4)), seed = 1)
sapply(J, rho_set, blocks = blocks, delta = 0.22)

## End(Not run)
```

information

Fisher Information and Risk Benchmarks

Description

`fisher_info` computes the observed-data Fisher information at the zero operator. `ipw_risk` computes the exact risk of the IPW estimator and the local asymptotic minimax constant.

Usage

```
fisher_info(rho, Sigma)
```

```
ipw_risk(rho, Gamma, Sigma, n)
```

Arguments

<code>rho</code>	co-observation probabilities of the directions.
<code>Sigma</code>	$d \times d$ error covariance.
<code>Gamma</code>	true coefficient matrix, with rows aligned with <code>rho</code> . Use truth_matrix to place the true coefficients in a dictionary.
<code>n</code>	sample size.

Details

Consider the Gaussian-error submodel with Rademacher scores. With the coefficients stacked by interaction and then by response coordinate, the information at $\Gamma = 0$ is

$$I(0) = \text{diag}(\rho_{J_1}, \dots, \rho_{J_M}) \otimes \Sigma^{-1}.$$

It is diagonal in the interaction index even when the sets overlap. This follows from Walsh orthogonality, not from disjointness. Directions with $\rho_J = 0$ have zero information because their scores vanish identically.

For resolvable directions, the local asymptotic minimax constant for squared loss is $\text{tr}\{I(0)^{-1}\} = \text{tr}(\Sigma) \sum_J \rho_J^{-1}$, and the IPW estimator attains it. The exact risk of IPW for Rademacher scores is

$$E\|\hat{g}_J - g_J\|^2 = \frac{1}{n} \left\{ \frac{\|\Gamma\|_F^2 + \text{tr}(\Sigma)}{\rho_J} - \|g_J\|^2 \right\}.$$

Adding the squared norms of trimmed coefficients gives the continuous reference risk used in the simulations. These are continuous-observation values; a fixed grid slightly reduces the numerical availability of scores.

Value

`fisher_info`: the $Md \times Md$ information matrix.

`ipw_risk`: a list with `exact_risk` (the risk of each direction, NA when $\rho_J = 0$), `lam_constant` ($\text{tr}(\Sigma) \sum \rho_J^{-1}$) and `lam_risk` (`lam_constant / n`).

See Also

[rbfit](#), [classify_sets](#)

Examples

```
## Not run:
## Design 6: information constant tr(V) = 28.917
st <- design_setup(6)
rho <- sapply(st$J, rho_set, blocks = st$blocks, delta = st$delta)
I0 <- fisher_info(rho, st$Sigma)
sum(diag(solve(I0)))
ipw_risk(rho, st$Gamma, st$Sigma, n = 1)$lam_constant

## Monte Carlo check at the null: mean of n * ||Gamma_hat||^2
## over the five directions (IPW is computed direction by direction)
keys <- sapply(st$J, paste, collapse = ",")
n <- 800
loss <- replicate(100, {
  d0 <- sim_design(6, n = n, alternative = "null")
  g <- coef(rbfit(d0, qmax = 2, tau = 0.01))[keys, ]
  n * sum(g^2)
})
mean(loss)

## local alternative: signed error along H is centred at zero
proj <- replicate(100, {
  dl <- sim_design(6, n = n, alternative = "local")
  g <- coef(rbfit(dl, qmax = 2, tau = 0.01))[keys, ]
  sqrt(n) * sum((g - st$H / sqrt(n)) * st$H) / sqrt(sum(st$H^2))
})
```

```

}))
mean(proj)

## Design 1: continuous IPW risks 1.478, 0.591 and 0.296
s1 <- design_setup(1)
cls <- classify_sets(s1$blocks, s1$delta, qmax = 2, tau = 0.01)
gt <- truth_matrix(cls$sets, s1$J, s1$Gamma)
sapply(c(200, 500, 1000), function(n)
  sum(ipw_risk(cls$rho, gt, s1$Sigma, n)$exact_risk[cls$rho >= 0.01]))

## End(Not run)

```

make_blocks

*Disjoint Block System of the Localized Basis***Description**

Builds the pairwise disjoint supports $I_1, \dots, I_K \subset (0, 1)$ of the localized predictor basis.

Usage

```
make_blocks(centers = c(0.075, 0.445, 0.845), per_cluster = 2,
            width = 0.03, gap = 0.06)
```

Arguments

centers	cluster centres in $(0, 1)$.
per_cluster	number of blocks in each cluster.
width	common block width.
gap	distance between neighbouring block centres within a cluster.

Details

Blocks in the same cluster are close together, so a short window can reveal them jointly. Blocks in different clusters are far apart, so no window of moderate length reveals both. The default gives the six supports used in the paper, with centres 0.045, 0.105, 0.415, 0.475, 0.815 and 0.875 and width 0.03.

An error is returned if two blocks overlap, because disjoint supports are required by the localized-score model.

Value

A list with the ordered lower and upper endpoints (lower, upper), the cluster of each block (cluster), and the number of blocks K.

See Also

[phi_basis](#), [resolution_order](#)

Examples

```
## Not run:
b <- make_blocks()
cbind(b$lower, b$upper)

## the four-block system of design 5
make_blocks(centers = c(0.14, 0.38, 0.62, 0.86), per_cluster = 1,
            width = 0.18, gap = 0.01)

## End(Not run)
```

parity_stat	<i>Visible Parity Statistic</i>
-------------	---------------------------------

Description

Computes a statistic from the second conditional moment that reveals the product of the signs of two overlapping non-resolvable interactions.

Usage

```
parity_stat(data, Jpair, component = 1)
```

Arguments

data	an object of class "rbdata".
Jpair	list of two index sets J_1 and J_2 .
component	response coordinate used.

Details

Non-resolvability ($\rho_J = 0$) removes a direction from the conditional mean, but not necessarily from the full observed law. Suppose $G = c(\sigma_1 W_{J_1} + \sigma_2 W_{J_2})$ with Rademacher scores, and the symmetric difference $J_1 \triangle J_2$ is resolvable. Then the conditional covariance of the response depends on $\sigma_1 \sigma_2$. The statistic

$$T_n = \frac{\sum_i B_i (Y_i^*)^2 \prod_{k \in J_1 \triangle J_2} \hat{\xi}_{ik}}{\sum_i B_i},$$

where B_i indicates that the symmetric difference is observed, converges to $2c^2 \sigma_1 \sigma_2$. The joint flip $(\sigma_1, \sigma_2) \rightarrow (-\sigma_1, -\sigma_2)$ leaves the observed law unchanged. In contrast, for disjoint families satisfying [hypercube_check](#), no such statistic exists, because all 2^m sign patterns give the same observed law.

Value

A list with statistic (T_n), sym_diff (the symmetric difference), rho (its co-observation probability) and n_seen (the number of subjects who observe it).

See Also

[hypercube_check](#), [sim_design](#)

Examples

```
## Not run:
## Design 7, overlapping family {1,3}, {1,2,3}: T_n -> 2 * sigma1 * sigma2
Jpair <- list(c(1, 3), c(1, 2, 3))
for (sg in list(c(1, 1), c(1, -1), c(-1, 1), c(-1, -1))) {
  dat <- sim_design(7, n = 30000, family = "overlap", signs = sg, seed = 1)
  cat(sg, " T_n =", round(parity_stat(dat, Jpair)$statistic, 3),
      " expected =", 2 * prod(sg), "\n")
}

## Design 7, hypercube family: the two antipodal vertices give the same law
for (sg in list(c(1, 1), c(-1, -1))) {
  dat <- sim_design(7, n = 1000, family = "hypercube", signs = sg, seed = 2)
  y <- dat$truth$Ystar[, 1]
  cat(sg, " mean =", round(mean(y), 3), " var =", round(var(y), 3), "\n")
}

## End(Not run)
```

predict.rbfit

Prediction from a Fragment

Description

Predicts the responses of new subjects from their own observed fragments, using a fitted "rbfit" object.

Usage

```
## S3 method for class rbfit
predict(object, newdata, ...)
```

Arguments

object	an object of class "rbfit".
newdata	an object of class "rbdata" with the new fragments.
...	not used.

Details

For a new subject, the prediction is $\hat{m} = \sum_J \hat{g}_J Z_J$. Here Z_J is the product of the recovered scores in J when the new window reveals all of them, and zero otherwise.

Directions that the new window does not reveal have conditional mean zero given the fragment, so the best fragment predictor omits them. As a consequence, prediction and operator recovery are different targets: a fit can predict well from fragments while leaving the non-resolvable part of the operator completely unknown.

Value

A list with components

Ystar	$n \times d$ matrix of predicted response coordinates.
Y	$n \times \text{length}(\text{tgrid})$ matrix of predicted response curves.

See Also

[rbfit](#), [evaluate_fit](#)

Examples

```
## Not run:
## Design 5: fully observed curves (delta = 1), Gaussian scores
train <- sim_design(5, n = 500, seed = 1)
test <- sim_design(5, n = 500, seed = 2)
fit <- rbfit(train, qmax = 1, tau = 0.01)
pr <- predict(fit, test)

## integrated squared error against the latent signal G(X)
truth <- test$truth$mu %*% t(test$E)
curve_mise(pr$Y, truth, test$tgrid)

## End(Not run)
```

rbdata

Fragmented Function-on-Function Data

Description

Collects fragmented predictor curves, response curves, window starts and the localized design into one object of class "rbdata". This object is the input of [rbfit](#).

Usage

```
rbdata(Xwin, Y, A, sgrid, tgrid, blocks, delta, d = 2, Phi = NULL, E = NULL)
```

Arguments

Xwin	$n \times \text{length}(\text{sgrid})$ matrix of predictor values, with NA outside each subject's window.
Y	$n \times \text{length}(\text{tgrid})$ matrix of response curves.
A	vector of window starting points, in $[0, 1 - \delta]$.
sgrid, tgrid	predictor and response grids on $[0, 1]$.
blocks	block system from make_blocks .
delta	window length.
d	number of response basis functions; used when E is NULL.
Phi	optional predictor basis; by default <code>phi_basis(blocks, sgrid)</code> .
E	optional orthonormal response basis; by default <code>response_basis(d, tgrid)</code> .

Details

Subject i contributes the observed datum $D_i = (A_i, M_{A_i} X_i, Y_i)$. The masking operator M_a keeps the predictor on $[a, a + \delta]$.

The fitting functions assume the following:

- the window start is independent of the predictor and the error;
- window starts are uniform on $[0, 1 - \delta]$; this is used to compute ρ_J for the IPW estimator;
- the localized system $\{\phi_k\}$ is known.

When $\delta = 1$, all curves are fully observed and A should be zero.

Value

An object of class "rbdata": a list with components `n`, `A`, `Xwin`, `Y`, `sgrid`, `tgrid`, `delta`, `blocks`, `Phi` and `E`.

See Also

[gen_fofr](#), [sim_design](#), [rbfit](#)

Examples

```
## Not run:
## wrap fragments produced elsewhere (here taken from a simulated sample)
sim <- sim_design(1, n = 300, seed = 1)
dat <- rbdata(Xwin = sim$Xwin, Y = sim$Y, A = sim$A,
             sgrid = sim$sgrid, tgrid = sim$tgrid,
             blocks = sim$blocks, delta = sim$delta, d = 2)
rbfit(dat, qmax = 2)

## End(Not run)
```

rbfit

Resolution-Aware Estimation of the Identifiable Component

Description

Estimates the Walsh coefficients g_J of all interactions of order at most `qmax` whose co-observation probability is at least `tau`. The remaining directions are set to zero. They are reported separately as trimmed or non-resolvable. This is Algorithm 2 of the paper.

Usage

```
rbfit(data, qmax = 2, tau = 0.01, method = c("ipw", "ratio"))

## S3 method for class rbfit
print(x, digits = 3, ...)

## S3 method for class rbfit
coef(object, ...)
```

Arguments

data	an object of class "rdata", created by <code>rbdata</code> , <code>gen_fofr</code> or <code>sim_design</code> .
qmax	largest interaction order in the dictionary.
tau	threshold on the co-observation probability; directions with $\rho_J \geq \tau$ are estimated.
method	"ipw" for the inverse-probability weighted estimator (known window law), or "ratio" for the complete-case ratio estimator (unknown window law).
x, object	an object of class "rbfit".
digits	number of digits printed.
...	not used.

Details

The dictionary contains all $J \subseteq \{1, \dots, K\}$ with $|J| \leq \text{qmax}$, including the empty set (the intercept). It is split into three groups: $J_\tau = \{\rho_J \geq \tau\}$, the trimmed set $\{0 < \rho_J < \tau\}$, and the non-resolvable set $\{\rho_J = 0\}$.

For each subject, the visible scores are recovered from the masked curve by $\xi_k = \langle M_a X, M_a \phi_k \rangle / \|M_a \phi_k\|^2$ (see `recover_all`). Invisible scores are never imputed. The response is represented by its coordinates Y^* on the response basis. With $\Delta_{iJ} = 1\{J \subseteq K(A_i)\}$, the two estimators are

$$\hat{g}_J = \frac{1}{n\rho_J} \sum_{i=1}^n \Delta_{iJ} Y_i^* W_J(X_i), \quad \tilde{g}_J = \frac{\sum_i \Delta_{iJ} Y_i^* W_J(X_i)}{\sum_i \Delta_{iJ}},$$

with $\tilde{g}_J = 0$ when no subject observes J .

The IPW estimator is unbiased for g_J , with risk $n^{-1}\{\rho_J^{-1}E\|Y\|^2 W_J^2 - \|g_J\|^2\}$. Near the zero operator it attains the local asymptotic minimax constant $\text{tr}(\Sigma) \sum_J \rho_J^{-1}$. The ratio estimator has the same local limit, but it is biased at finite n by the factor $1 - (1 - \rho_J)^n$.

The truncated fit satisfies the exact decomposition

$$E\|\hat{G}_\tau - G\|^2 = \sum_{J \in J_\tau} E\|\hat{g}_J - g_J\|^2 + \sum_{0 < \rho_J < \tau} \|g_J\|^2 + \sum_{\rho_J = 0} \|g_J\|^2.$$

The three terms are the stochastic error, the threshold bias, and the energy absent from the conditional mean. Only the first term decreases with n .

Because each coefficient is estimated separately, the estimate of a given direction does not depend on which other directions are in the dictionary.

Value

An object of class "rbfit", a list with components:

coefficients	matrix of estimated coefficients, one row per dictionary set (row names are set keys, with "{}" for the intercept) and one column per response basis function.
method	the estimator used.
sets, keys	the dictionary, as a list of index sets and as keys.
rho	co-observation probabilities of the dictionary sets.
kept	logical, $\rho \geq \tau$.
report	list with the keys of the estimated, trimmed and nonresolvable directions.

p resolution order.
 qmax, tau, delta, n settings and sample size.
 xihat, D, features recovered scores, observability indicators and observed Walsh features.
 coef returns the coefficient matrix. print shows a summary and the estimated coefficients.

See Also

[predict.rbfit](#), [evaluate_fit](#), [classify_sets](#), [ipw_risk](#)

Examples

```
## Not run:
## Design 1: all active directions are resolvable
train <- sim_design(1, n = 1000, seed = 1)
fit <- rbfit(train, qmax = 2, tau = 0.01, method = "ipw")
fit
coef(fit)
fit$report$trimmed

## unknown window law: complete-case ratio estimator
fit_ratio <- rbfit(train, qmax = 2, tau = 0.01, method = "ratio")

## Design 3 at delta = 0.22: the cubic terms {1,2,3} and {4,5,6} are
## non-resolvable and are reported, not estimated
dat <- sim_design(3, n = 1000, delta = 0.22, seed = 2)
fit3 <- rbfit(dat, qmax = 3, tau = 0.01)
fit3$p
c("1,2,3", "4,5,6") %in% fit3$report$nonresolvable

## Design 2: the same sample under the threshold grid of the paper
dat2 <- sim_design(2, n = 500, seed = 3)
st <- dat2$setup
for (tau in st$extra$tau_grid) {
  f <- rbfit(dat2, qmax = 2, tau = tau)
  err <- operator_error(f, st$J, st$Gamma)
  cat("tau =", tau, " kept =", sum(f$kept),
      " loss =", round(err$total, 3),
      " threshold bias =", round(err$threshold, 3), "\n")
}

## End(Not run)
```

recover_all

Recover Visible Scores from Fragments

Description

Recovers, for each subject, the scores of the blocks revealed by the subject's window.

Usage

```
recover_all(data)
```

Arguments

data an object of class "rdata".

Details

For $k \in K(a)$, the score is recovered exactly by the projection

$$\xi_k = \frac{\langle M_a X, M_a \phi_k \rangle}{\|M_a \phi_k\|^2}.$$

The denominator is positive because $\phi_k > 0$ inside its support, and the disjoint supports make the projection exact.

On a grid, the inner products are computed by the trapezoidal rule. A score is declared unavailable when the squared norm is at most 10^{-10} . Consequently, a block that is visible in continuous time can occasionally be unavailable numerically, and the numerical availability satisfies $\eta_J \leq \rho_J$. Scores of blocks that are not revealed are returned as NA; they are never imputed.

Value

An $n \times K$ matrix of recovered scores, with NA where a block is not visible.

See Also

[rbfit](#)

Examples

```
## Not run:
## Design 4: recovery error on visible scores is at round-off level
dat <- sim_design(4, n = 500, seed = 1)
xi <- recover_all(dat)
vis <- !is.na(xi)
max(abs(xi[vis] - dat$truth$xi[vis]))
colMeans(vis) # proportion of subjects seeing each block

## End(Not run)
```

resolution_audit

Resolution Audit of a Uniform Window Design

Description

For each interaction order $r = 1, \dots, Q$, reports the volume of tuples whose co-observation probability is at least tau, next to the unthresholded band volume. It then returns the largest order that meets a coverage criterion.

Usage

```
resolution_audit(delta, Q = 5, tau = 0.05, floor_prop = 0.02,
  nmc = 20000, seed = NULL)
```

Arguments

delta	window length.
Q	largest order examined.
tau	threshold on the co-observation probability.
floor_prop	minimum thresholded volume for an order to be counted as supported.
nmc	number of Monte Carlo points per order.
seed	optional random seed.

Details

Under uniform windows of length δ , the function estimates the volume $V_r(\tau) = \lambda_r\{s : \rho_r(s) \geq \tau\}$ by Monte Carlo. It returns the volume for each order together with the band volume $r\delta^{r-1} - (r-1)\delta^r$, and r^* , the largest order whose thresholded volume is at least floor_prop.

The thresholded volume has the closed form $V_r(\tau) = rLb^{r-1} - (r-1)b^r$. Here $t = \tau(1 - \delta)$, $L = 1 - 2t$ and $b = \delta - t$. This closed form can be used to check the Monte Carlo value; at $\delta = 0.30$ and $\tau = 0.05$ it gives 0.4227, 0.1587, 0.0544 and 0.0177 for orders 2 to 5. A positive threshold lowers the volume even when all windows have the same length.

The value r^* describes direct co-observation under the chosen threshold. It does not say which orders are identified. Identification is decided by ρ_J and the resolution order (see [classify_sets](#)). This function audits the controlled uniform design; the empirical interval audit of the paper (Algorithm 1) works from recorded interval endpoints.

Value

A list with components table (a data frame with columns order, reachable and band), r_star, tau and floor_prop.

See Also

[band_measure](#), [classify_sets](#)

Examples

```
## Not run:
aud <- resolution_audit(delta = 0.30, Q = 5, tau = 0.05,
                        nmc = 200000, seed = 1)

aud$table
aud$r_star

## closed-form benchmark for comparison
t0 <- 0.05 * (1 - 0.30); L <- 1 - 2 * t0; b <- 0.30 - t0
sapply(2:5, function(r) r * L * b^(r - 1) - (r - 1) * b^r)

## End(Not run)
```

sim_design

*Simulate Data from the Seven Simulation Designs***Description**

Generates one sample from any of the seven simulation designs of the paper.

Usage

```
sim_design(design, n, delta = NULL,
           alternative = c("null", "local"),
           family = c("hypercube", "overlap"),
           signs = c(1, 1), seed = NULL)
```

Arguments

design	integer from 1 to 7; see design_setup .
n	sample size.
delta	window length; used only in design 3.
alternative	design 6 only: "null" for the zero operator, or "local" for the local alternative $H/\sqrt{n}$.
family	design 7 only: "hypercube" for the disjoint cubic family {1,2,3}, {4,5,6}, or "overlap" for {1,3}, {1,2,3}.
signs	design 7 only: the signs (σ_1, σ_2) .
seed	optional random seed.

Details

The data are generated by [gen_fofr](#). Training and test samples are obtained by calling the function twice with different seeds. The settings actually used, including the true operator, are stored in setup.

Value

An object of class "rdata" with the extra components truth (see [gen_fofr](#)) and setup (see [design_setup](#)). The true index sets and coefficients are setup\$J and setup\$Gamma.

See Also

[design_setup](#), [rbfit](#), [evaluate_fit](#)

Examples

```
## Not run:
## Design 1: resolvable nonlinear effects
train <- sim_design(1, n = 500, seed = 1)
test <- sim_design(1, n = 1000, seed = 2)
fit <- rbfit(train, qmax = 2)
evaluate_fit(fit, test, train$setup$J, train$setup$Gamma)

## Design 3: operator loss and fragment-signal loss across window lengths
```

```
for (dl in c(0.22, 0.38, 1)) {  
  tr <- sim_design(3, n = 1000, delta = dl, seed = 3)  
  te <- sim_design(3, n = 1000, delta = dl, seed = 4)  
  f <- rbfit(tr, qmax = 3)  
  print(round(evaluate_fit(f, te, tr$setup$J, tr$setup$Gamma)[  
    c("operator_total", "operator_nonresolvable", "fragment_signal_mse")], 3))  
}  
  
## Design 4: Gaussian scores; the omitted cubic energy 2.5 remains  
dat <- sim_design(4, n = 2000, seed = 5)  
err <- operator_error(rbfit(dat, qmax = 3), dat$setup$J, dat$setup$Gamma)  
c(total = err$total, estimated = err$estimated,  
  omitted = err$nonresolvable)  
  
## Design 6: local alternative  $H / \sqrt{n}$   
dat <- sim_design(6, n = 1600, alternative = "local", seed = 6)  
  
## Design 7: overlapping family with opposite signs  
dat <- sim_design(7, n = 30000, family = "overlap",  
  signs = c(1, -1), seed = 7)  
  
## End(Not run)
```

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